

BIM308 – Web Server Programming – 4th Assignment

In this homework, you will develop a University Information System using a **RESTful API**. This system will include key components such as Students, Courses, and Classrooms. The API will facilitate interaction between these components using **HTTP methods** (GET, POST, PUT, DELETE).

1. Sample Data Models (JSON Format)

Author:

```
{
  "authorID": 1,
  "authorName": " George Orwell",
  "authorInfo": " English novelist and essayist",
},
{
  "authorID": 2,
  "authorName": " J.K. Rowling",
  "authorInfo": " British author, Harry Potter series",
},
...
```

User:

```
{
  "userID": 1,
  "name": "Emin Demirkiran",
  " email": "emin@mail.com",
  " rentedBookIDs": [1, 3, 5]
},
{
  "userID": 2,
  "name": "Elif Şahin",
  " email": "elif@mail.com",
  " rentedBookIDs": [6, 10]
}
...
```

Book:

```
{
  "bookID": 1,
  "title": "1984",
  "releaseYear": "1949",
  "price": 120,
  "authorID": 1,
},
{
  "bookID": 2,
  "title": "Animal Farm",
  "releaseYear": "1945",
  "price": 90,
  "authorID": 1,
}
...
```

Sample JSON files prepared for your use: https://github.com/emindk/JSON_Files

2. System Relationships (Conceptual)

- A Book belongs to an Author
- A User can rent multiple Books
- rentedBookIDs field represents the books borrowed by users

3. Expected RESTful API Functionalities

Function	HTTP Method	Endpoint	Description
List all authors	GET	/api/authors/	Retrieve all authors
Get a specific author	GET	/api/authors/{id}	Retrieve author details
Add a new author	POST	/api/authors	Create a new author
Update author	PUT	/api/authors/{id}	Update author info
Delete author	DELETE	/api/authors/{id}	Remove an author
List all books	GET	/api/books	Retrieve all books
Get a specific book	GET	/api/books/{id}	Retrieve book details
Add a new book	POST	/api/books	Create a new book
Update book	PUT	/api/books/{id}	Update book info
Delete a book	DELETE	/api/books/{id}	Remove a book
List all users	GET	/api/users	Retrieve all users
Get a specific user	GET	/api/users/{id}	Retrieve user details
Add a new user	POST	/api/users	Create a new user
Update user	PUT	/api/users/{id}	Update user info
Delete user	DELETE	/api/users/{id}	Remove a user
Rent a book	POST	/api/users/{userId}/rent/{bookId}	User rents a book
Return a book	POST	/api/users/{userId}/return/{bookId}	User returns a book

4. Technical Requirements

- RESTful API principles must be followed
- Proper HTTP status codes should be used (200, 201, 404, 400, etc.)
- JSON format must be used for request/response
- Layered architecture is recommended:
 - Controller
 - Service
 - Repository (optional)

Submitting the Assignment

Zip up your project which should be named as **yourlastname_yourID_hw4.zip** and submit it to ESTUOYS platform until May 10 at 23:59.

Only one group member should make submission. Also, please provide a **ReadMe.txt** file that gives information about the group members. Otherwise, your colleague might not receive any points.

Do not forget to put your Project into a container by using the **Dockerfile** that is given to you. Be careful! You will need to modify your Dockerfile and change the last line from "**BookLibrary.dll**" to "**BookLibraryREST.dll**".

Late submission is not allowed by the system. Submissions via e-mail will not be accepted by the instructors and they will be deleted immediately.